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I. System Overview:

The Papaya Beach Resort Reservation and Activity Management System is an integrated web-based platform that enables streamlined booking and management of resort accommodations and optional water activities. The system is designed to handle guest reservations, online payments, and administrative tasks to ensure efficient resort operations. It includes three main user roles: Guest, Staff, and Admin, each with specific functionalities suited to their responsibilities. The system ensures a seamless experience from room search and booking to check-out, with added support for optional activities and automated reservation confirmation.

II. Business Problem:

Papaya Beach Resort faces challenges in managing increasing guest reservations, activity bookings, and payment tracking using traditional manual processes. These methods result in booking conflicts, delayed confirmations, data loss, and poor customer service. There is also a lack of real-time availability tracking and no centralized system for staff and administrative oversight. The developed system addresses these problems by providing a centralized, digital platform for managing all reservation-related tasks, enhancing guest experience and internal efficiency.

III. Introduction

Papaya Beach Resort, located in a scenic tourist destination, offers lodging and boat activities for families and groups. The resort caters to hundreds of guests every month, especially during peak seasons. Previously relying on manual booking ledgers and spreadsheets, the resort found it increasingly difficult to keep up with guest demands, track payments, and manage boat schedules. The shift to a digital reservation system is aimed at enhancing operational efficiency, ensuring accurate records, and delivering faster service.

IV. Objectives

- Automate the process of guest account creation and secure login.
- Allow guests to search and view room availability based on preferred dates and number of occupants.
- Enable real-time room selection with editable reservation details and optional water activity booking.
- Support scheduling of water activities within the check-in to check-out window.
- Provide a booking summary with 30% downpayment calculation and online payment integration.
- Process QR code-based payments and accept guest reference numbers for verification.
- Allow guests to manage pending and confirmed reservations and update personal information.
- Empower staff to manage and confirm bookings, monitor payments, and handle check-in/check-out operations.



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- Enable administrators to manage system users, rooms, boats, and reservation records, and view analytics.

V. Data Sources and Preparation

○ Data Sources

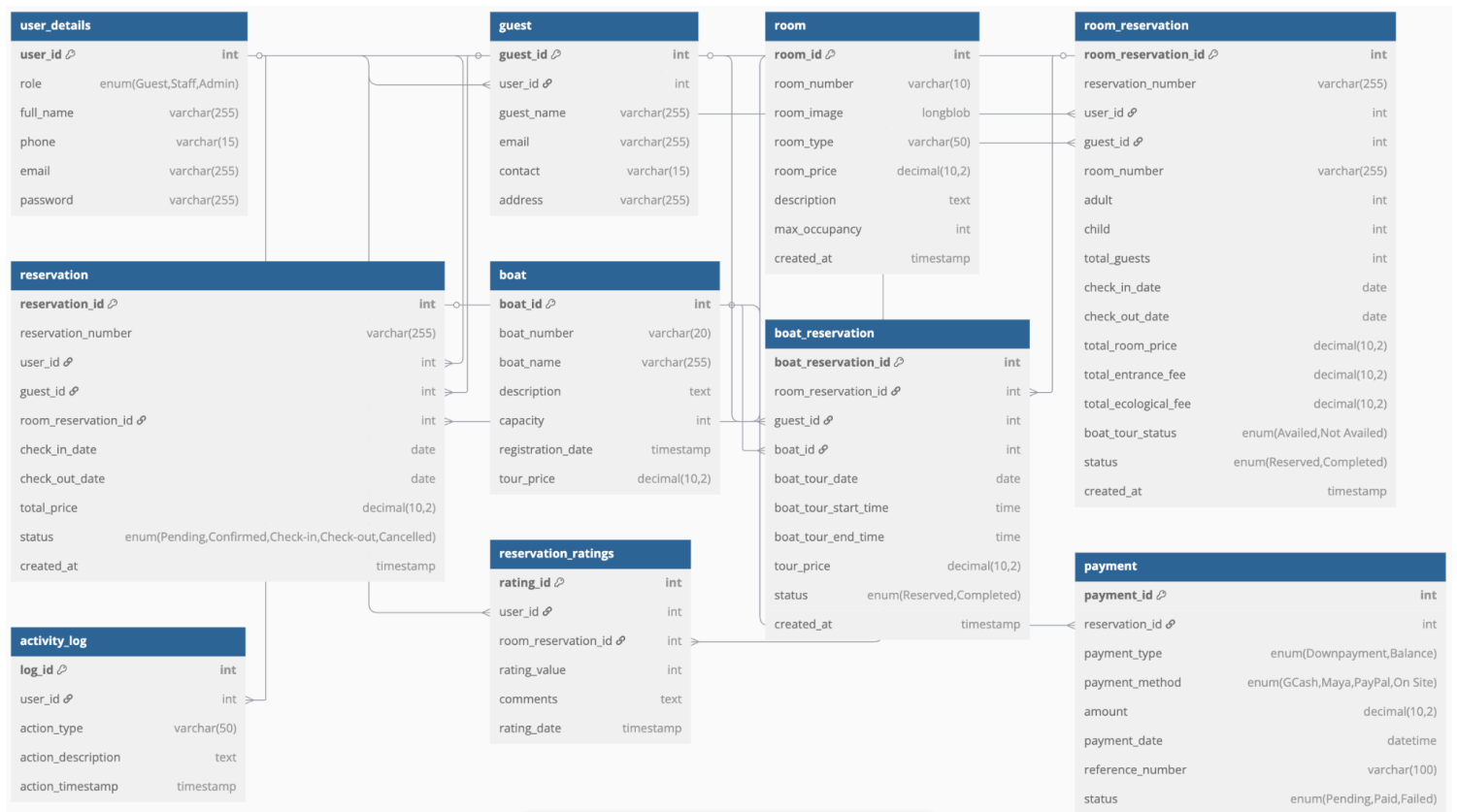


Figure 1. Entity Relationship Diagram

○ Table and Relationship

Table user_details {
user_id int [pk]
role enum('Guest', 'Staff', 'Admin')
full_name varchar(255)
phone varchar(15)
email varchar(255)
password varchar(255) }

Table guest {
guest_id int [pk]
user_id int [ref: > user_details.user_id]
guest_name varchar(255)
email varchar(255)
contact varchar(15)
address varchar(255)}

contact varchar(15)
address varchar(255)}

Table room {
room_id int [pk]
room_number varchar(10)
room_image longblob
room_type varchar(50)
room_price decimal(10,2)
description text
max_occupancy int
created_at timestamp}



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```
Table room_reservation {
  room_reservation_id int [pk]
  reservation_number varchar(255)
  user_id int [ref: > user_details.user_id]
  guest_id int [ref: > guest.guest_id]
  room_number varchar(255)
  adult int
  child int
  total_guests int
  check_in_date date
  check_out_date date
  total_room_price decimal(10,2)
  total_entrance_fee decimal(10,2)
  total_ecological_fee decimal(10,2)
  boat_tour_status enum('Availed', 'Not Availed')
  status enum('Reserved', 'Completed')
  created_at timestamp }
```

```
Table reservation {
  reservation_id int [pk]
  reservation_number varchar(255)
  user_id int [ref: > user_details.user_id]
  guest_id int [ref: > guest.guest_id]
  room_reservation_id int [ref: >
room_reservation.room_reservation_id]
  check_in_date date
  check_out_date date
  total_price decimal(10,2)
  status enum('Pending', 'Confirmed', 'Check-in',
'Check-out', 'Cancelled')
  created_at timestamp }
```

```
Table boat {
  boat_id int [pk]
  boat_number varchar(20)
  boat_name varchar(255)
  description text
  capacity int
  registration_date timestamp
  tour_price decimal(10,2)}
```

```
Table boat_reservation {
  boat_reservation_id int [pk]
```

```
  room_reservation_id int [ref: >
room_reservation.room_reservation_id]
  guest_id int [ref: > guest.guest_id]
  boat_id int [ref: > boat.boat_id]
  boat_tour_date date
  boat_tour_start_time time
  boat_tour_end_time time
  tour_price decimal(10,2)
  status enum('Reserved', 'Completed')
  created_at timestamp }
```

```
Table payment {
  payment_id int [pk]
  reservation_id int [ref: >
reservation.reservation_id]
  payment_type enum('Downpayment', 'Balance')
  payment_method enum('GCash', 'Maya', 'PayPal',
'On Site')
  amount decimal(10,2)
  payment_date datetime
  reference_number varchar(100)
  status enum('Pending', 'Paid', 'Failed')}
```

```
Table activity_log {
  log_id int [pk]
  user_id int [ref: > user_details.user_id]
  action_type varchar(50)
  action_description text
  action_timestamp timestamp }
```

```
Table reservation_ratings {
  rating_id int [pk, increment]
  user_id int [ref: > user_details.user_id]
  room_reservation_id int [ref: >
room_reservation.room_reservation_id]
  rating_value int
  comments text
  rating_date timestamp
}
```




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- **Data preparation**
- **Data Cleaning**
 - Removed duplicate entries in guest, room, and reservation records to maintain data accuracy.
 - Enforced constraints such as NOT NULL, UNIQUE, and FOREIGN KEY to ensure relational integrity across tables.
- **Aggregation**
 - Calculated the total number of bookings, cancellations, and check-ins per room and boat for given time ranges.
 - Aggregated income data by summing up payments from room and boat reservations.
- **Join Operations**
 - Used INNER JOIN to combine data from reservation, guest, room, and boat tables for detailed reservation records.
 - Used LEFT JOIN to display all room or boat listings, including those without any current reservation.
- **SQL Views**
 - Created views to present real-time information such as today's check-ins, active reservations, and available rooms or boats.
 - Simplified reporting and dashboard queries by using views for total revenue, guest statistics, and occupancy rates.
- **Stored Procedures**
 - Developed procedures for frequently used operations like retrieving available rooms, checking guest history, and computing total payments.
 - Used procedures to encapsulate revenue calculations and automate reservation validation.
- **CRUD Operations:**
 - Implemented Create, Read, Update, and Delete functionalities for guests, rooms, boats, reservations, and payments.
 - Enabled user-friendly management of records through Java Swing forms integrated with MySQL.
- **Events:**
 - Scheduled automatic cleanup of expired or cancelled reservations to improve database performance.
 - Set events for generating daily summary reports or reminders for upcoming guest check-ins.
- **Triggers:**
 - Configured triggers to update room and boat availability upon reservation insertion or cancellation.



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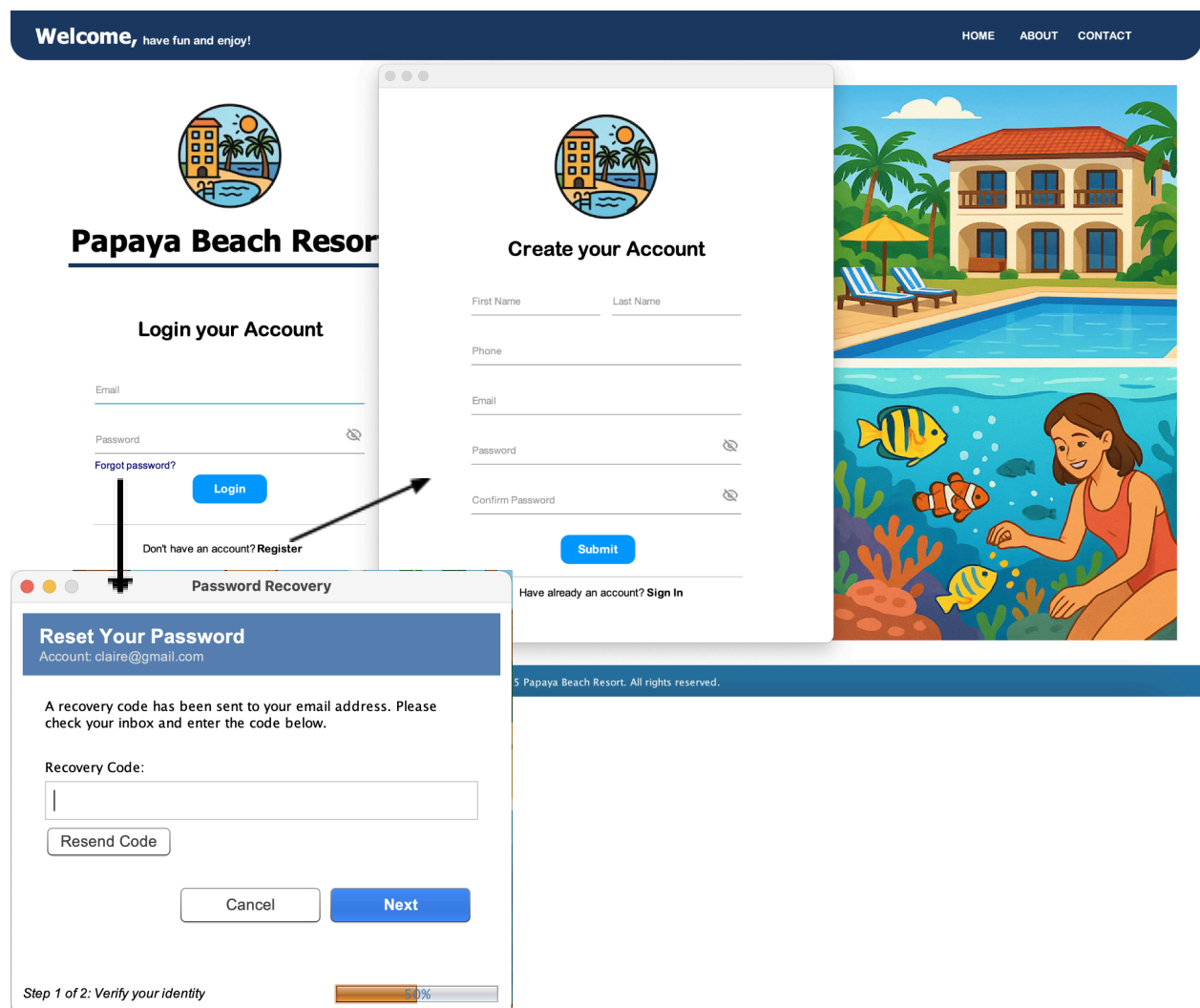
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- Set triggers to log changes in reservation status (e.g., from “Reserved” to “Checked-in”) for audit trail purposes.

VI. Screenshots

- Guest Interface



This panel allows new guests to register by entering a valid email and a secure password. Returning users can log in through this form, where the system validates credentials against the database. If users forget their password, they can click the **"Forgot Password"** link, where the system will send a confirmation code to their registered email address. After entering the correct code, they will be allowed to reset their password securely.



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Welcome, have fun and enjoy!

HOMEABOUTCONTACT

About Us



Welcome to Paradise

Established in 2010, **Papaya Beach Resort** has been providing unforgettable beach experiences for over 15 years. Our luxurious beachfront property spans 12 acres of pristine coastline, offering breathtaking views and direct access to crystal-clear waters and white sandy beaches.

We pride ourselves on offering **world-class accommodations** paired with **exceptional service**. Our dedicated staff ensures that each guest experiences the perfect blend of relaxation and adventure during their stay.

From thrilling water activities to serene spa treatments, we cater to all types of travelers seeking a memorable beach getaway. Our commitment to sustainability ensures that the natural beauty of our surroundings is preserved for generations to come.

WATER ACTIVITIES

Explore the Best Aquatic Adventures

© 2025 Papaya Beach Resort. All rights reserved.

This section highlights Papaya Beach Resort’s reputation for exceptional beachfront experiences since 2010. Set on 12 acres of pristine coastline, the resort offers world-class accommodations, stunning views, and direct beach access. It showcases a perfect blend of relaxation and adventure, from water activities to serene spa treatments, all with a strong commitment to sustainability.

Welcome, have fun and enjoy!

HOMEABOUTCONTACT

How Can We Help You?

We'd love to hear from you

Contact Information

Phone

+1 (555) 123-4567
24/7 Customer Support
For emergencies: +1 (555) 987-6543

Email

papayabeach@gmail.com
General inquiries
papayarreservation@gmail.com
Booking requests

Location

Papaya Beach Resort
St. Dagundong, Brgy. Papaya,
Nasugbu, Batangas

View Location

Hours

Front Desk: 24/7
Dining: 7AM-11PM
Spa: 9AM-8PM
Pool: 6AM-10PM

Send Us a Message

Have questions or special requests? We'll respond within 24 hours.

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This section offers contact details, location with a "View Location" button, service hours, and a form for guest inquiries with a 24-hour response time. Here, guests input their desired check-in and check-out dates along with the number of adults and children. The system processes this to show which rooms are available for their selected criteria.

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PAPAYA BEACH RESORT, *Escape to Paradise*

HOMERESERVATIONPROFILELog out



Reserve your dates

Check-in Date

July 26, 2025



Check-out Date

July 28, 2025



Adult

2



Child

1



SEARCH

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Here, guests input their desired check-in and check-out dates along with the number of adults and children. The system processes this to show which rooms are available for their selected criteria.

PAPAYA BEACH RESORT, *Escape to Paradise*
HOME
RESERVATION
PROFILE
[Logout](#)

SEARCH YOUR ROOM

Available rooms for 3 guests from July 26, 2025 to July 28, 2025

Family Suite ★★★★☆ 4.2 (5) Max Guests: 6 Large suite with two double beds, sofa, dining space, and kitchenette. BOOK NOW • ₱5,000.00	Cabana ★★★★★ 4.0 (1) Max Guests: 4 Open-air cabana with native décor, hammock, and fan-cooled environment. BOOK NOW • ₱3,800.00	Beachfront Villa ★★★★★ 0.0 (0) Max Guests: 6 Beachfront villa with modern amenities and patio hammock. BOOK NOW • ₱9,100.00	Beachfront Villa ★★★★★ 0.0 (0) Max Guests: 6 Private villa steps from the sand with personal pool. BOOK NOW • ₱8,900.00
---	--	--	--

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This panel displays all available rooms matching the guest's input. Guests can change the check-in/check-out dates and number of guests if needed. Upon selection, the system prompts whether they want to avail of water activities. If yes, the activity scheduling form appears.



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PAPAYA BEACH RESORT, *Escape to Paradise*

HOME

RESERVATION

PROFILE

Logout

Date

Jul 26, 2025 (Sat)

Time

01:55 PM

SEARCH

Select a date

Jul 26, 2025 (Sat)

Jul 27, 2025 (Sun)

Jul 28, 2025 (Mon)

Available Boat Tours

For Jul 26, 2025 (Sat) at 01:55 PM

Ocean Spirit

₱1000.00 per ride

Spacious and luxurious ride.

SELECT

Reef Runner

₱1400.00 per ride

Compact boat for reef trips.

SELECT

Island Hopper

₱1500.00 per ride

Perfect for small island tours.

SELECT

Tide Chaser

₱1500.00 per ride

SELECT

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This panel is only shown if the guest opts for water activities. It allows selection of activity dates (within the stay duration) and time slots between 7:00 AM and 2:00 PM.

PAPAYA BEACH RESORT, *Escape to Paradise*

HOME

RESERVATION

PROFILE

Logout

Lead Guest Details

First Name

Claire

Last Name

Cabral

Email

claire@gmail.com

Contact Number

09123456780

Address Details

Region

REGION IV-A (CALABAR...

Province

BATANGAS

Municipal

NASUGBU

Barangay

LUMBANGAN

Proceed to Payment

Digital Payment

GCash

You have chosen to pay by GCash. You will be forwarded to the GCash website to proceed with this transaction. The total amount to be pay is: ₱2808.00

Pay with GCash

Sat, Jul 26

>>>

Mon, Jul 28

2

Check-in

Check-out

Nights

Number: R004

Room Type: Cabana

Inclusion:

Description

Open-air cabana with native decor, hammock, and fan-cooled environment.

Reef Runner

Sat, Jul 26

01:55 PM

04:55 PM

Boat Name

Date

Start Time

End Time

Reservation Number: RES-20250517-001

Guest Count: 2 Adults, 1 Child

Total Room Price

₱7600.00

Water Activity Price

₱1400.00

Others:

Total Entrance Fee

₱300.00

Total Ecological Fee

₱60.00

Total Price

₱9360.00

Downpayment

₱2808.00

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This panel summarizes all selected services (room + activity). It calculates 30% downpayment from the total cost. Guests must then enter the lead guest’s and address details and choose a payment method (GCash, Maya, or PayPal).



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My Reservation

Pending Reservation

Rate Your Stay

How was your stay?

★★★★★

Share your experience (optional):

Cancel Submit Rating

Check-out: Apr 26, 2025
Room: R004

View Details Rate Now

Guests can view two sections on their reservation page: 'Pending Reservations' for bookings awaiting confirmation with the option to cancel, and 'Active Reservations' for confirmed bookings where guests can rate their room experience up to 5 stars."

My Account

Full Name
Claire Cabral

Phone
09123456780

Email
claire@gmail.com

Update

Guests can view and update their personal information, such as name, contact number, and email address.



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Staff Interface

PAPAYA
Beach Resort

DASHBOARD

ROOMS

BOATS

PENDING

CHECK IN/OUT

RESERVATION

WELCOME, Clarisse Cabral

Logout

Beach-Front Resort Management System

Available Room 16

Reserve Room 48

Confirm Reservation 12

Pending Reservation 7

Available Boat 10

Reserve Boat 28

Cancel Reservation 5

Total Guest 17

Upcomming Check-In

17 May 2025 01:00:02

Reservation Nu...	Guest Name	Check-In Date	Check-Out Date	Total Price	Room Number	Boat Tour Status
RES-20250513-001	Jake Estanilo	2025-05-22	2025-05-23	4320.00	R010	Avalied
RES-20250516-003	Aldrie Baquiran	2025-05-22	2025-05-23	5240.00	R002	Not Avalied
RES-20250511-003	Jake Estanilo	2025-05-22	2025-05-23	9240.00	R003	Avalied
RES-20250515-001	Lei Torres	2025-05-29	2025-05-30	7040.00	R008	Avalied
RES-20250513-004	Yeej Valdez	2025-05-30	2025-05-31	3540.00	R015	Not Avalied
RES-20250511-005	Aj Catili	2025-06-25	2025-06-26	6880.00	R002	Avalied
RES-20250512-001	Aldrie Baquiran	2025-06-27	2025-06-28	5140.00	R001	Avalied
RES-20250511-002	Claire Cabral	2025-07-16	2025-07-18	10360.00	R002	Not Avalied
RES-20250511-001	Anthony Baquiran	2025-08-02	2025-08-05	15360.00	R002	Not Avalied
RES-20250511-006	Marie Santos	2025-08-19	2025-08-20	5440.00	R004	Avalied
RES-20250512-004	Lovely Masangkay	2025-10-23	2025-10-24	6240.00	R002	Avalied

This panel gives resort staff a quick overview of key operations. It displays total counts of available and reserved rooms/boats, and tracks reservation statuses (pending, confirmed, cancelled). It also features an “Upcoming Check-In” section showing guest names, dates, room numbers, and tour status. A real-time clock and date help with scheduling, and a sidebar allows navigation to other staff tools like room, boat, and reservation management.

List of Rooms

View Reservation

All

Search here...

Room Number	Room Image	Type	Description	Max Occupancy	Price/
R003		Beachfront Villa	Direct access to the beach, king-size bed...	2	7000
R004		Cabana	Open-air cabana with native décor, ham...	4	3800
R005		Premium Suite	Elegant suite with a king-size bed, bath...	2	6200
R006		Cabana	A cozy, private room typically near the b...	2	10000
R007		Deluxe Room	Modern deluxe room with sea breeze an...	2	3500
R008		Family Suite	Family suite with bunk beds and beachfr...	5	5400

Reservation Date

Reservation Number	Check-in Date	Check-out Date
RES-20250426-001	2025-04-25	2025-04-26
RES-20250426-002	2025-04-25	2025-04-26
RES-20250426-003	2025-04-25	2025-04-26
RES-20250426-004	2025-04-25	2025-04-26
RES-20250426-005	2025-04-25	2025-04-26
RES-20250426-006	2025-04-28	2025-04-29
RES-20250426-007	2025-04-12	2025-04-13
RES-20250427-001	2025-04-03	2025-04-04
RES-20250427-002	2025-04-04	2025-04-05
RES-20250427-003	2025-04-26	2025-04-27

This panel displays all rooms in the system with images and their reservation dates. It helps staff track occupancy and availability, allowing for efficient scheduling and avoiding booking conflicts."



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List of Boats

View Reservation

All

Search here...

Boat Number	Name	Details	Capacity	Price
BN-001	Island Hopper	Perfect for small island tours.	6	150
BN-002	Sunset Cruiser	Ideal for sunset cruises and private trips.	8	200
BN-003	Wave Rider	Fast boat for adventure seekers.	4	180
BN-004	Coral Explorer	Great for snorkeling and coral viewing.	10	250
BN-005	Sea Breeze	Comfortable boat for families.	7	170
BN-006	Lagoon Drifter	Relaxing tour through calm lagoons.	5	160
BN-007	Ocean Spirit	Spacious and luxurious ride.	12	100
BN-008	Reef Runner	Compact boat for reef trips.	4	140
BN-009	Blue Horizon	Good for medium group tours.	9	220
BN-010	Tide Chaser	Exciting speedboat for thrill rides.	7	150

Reservation Date

Reservation	Date	Start Time	Return Time
RES-20250426-001	2025-04-26	09:30 AM	12:30 PM
RES-20250426-002	2025-04-26	01:47 AM	04:47 AM
RES-20250426-003	2025-04-26	09:52 AM	12:52 PM
RES-20250426-004	2025-04-26	03:06 PM	06:06 PM
RES-20250426-005	2025-04-26	04:17 PM	07:17 PM
RES-20250426-006	2025-04-29	04:17 PM	07:17 PM
RES-20250427-004	2025-04-17	04:31 AM	07:31 AM
RES-20250502-001	2025-05-21	01:40 PM	04:40 PM
RES-20250502-002	2025-05-16	01:52 PM	04:52 PM
RES-20250510-002	2025-05-09	10:32 AM	01:32 PM

This panel displays all boats in the system along with their reservation dates. It helps staff monitor availability and manage scheduling efficiently to avoid booking conflicts."

Reservation

Search here...

Reservation Number	Guest Name	Check-In Date	Check-Out Date	Total Price	Created At	Actions
RES-20250513-002	Zeth Pagkaliwagan	2025-			2025-05-14 01:09:53	View Confirm Reject
RES-20250513-005	Zeth Pagkaliwagan	2025-			2025-05-14 02:01:07	View Confirm Reject
RES-20250513-006	Aeron Ramos	2025-			2025-05-14 02:06:34	View Confirm Reject
RES-20250516-001	Aldrie Baquiran	2025-			2025-05-17 08:54:23	View Confirm Reject
RES-20250516-002	Aldrie Baquiran	2025-			2025-05-17 08:55:12	View Confirm Reject
RES-20250516-004	Jake Stanio	2025-			2025-05-17 09:06:43	View Confirm Reject
RES-20250517-001	Claire Cabral	2025-			2025-05-17 15:34:00	View Confirm Reject

Reservation Details

Reservation Status: Pending

Downpayment Details:

Amount: ₱4344.00

Payment Method: PayPal

Reference Number: 1234567890

Payment Status: Pending

OK

This panel lists all pending reservations. Staff can view complete booking details, including the payment reference number, to confirm valid reservations or reject invalid ones.

Check-in

Check-out

Search here...

Reservation Number	Guest Name	Check-In Date	Check-Out Date	Total Price	Created At	Actions
RES-20250511-001	Anthont Baquiran	2025-08-02	2025-08-05	₱15360.00	2025-05-12 01:29:13	View Check-in
RES-20250511-002	Claire Cabral	2025-07-16	2025-07-18	₱10360.00	2025-05-12 09:08:00	View Check-in
RES-20250511-003	Jake Estanio	2025-05-22	2025-05-23	₱9240.00	2025-05-12 10:02:05	View Check-in

Here, staff can process guest check-in. It shows guest and reservation details, and the system calculates the remaining balance. Once paid, the system marks the booking as "Checked-In."



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Check-in

Check-out

Search here...

Reservation Number	Guest Name	Check-In Date	Check-Out Date	Total Price	Created At	Actions	
RES-20250517-004	Lovely Masangaky	2025-05-17	2025-05-19	P18800.00	2025-05-17 16:28:00	View	Check-out
RES-20250517-005	Joseph Tercias	2025-05-17	2025-05-18	P6240.00	2025-05-17 16:29:44	View	Check-out
RES-20250517-006	Zeth Pagkaliwagan	2025-05-17	2025-05-18	P6640.00	2025-05-17 16:30:37	View	Check-out

This panel is used during guest departure. It finalizes the reservation, logs feedback (optional), and updates the status to "Checked-Out."

Completed Reservation

Search here..

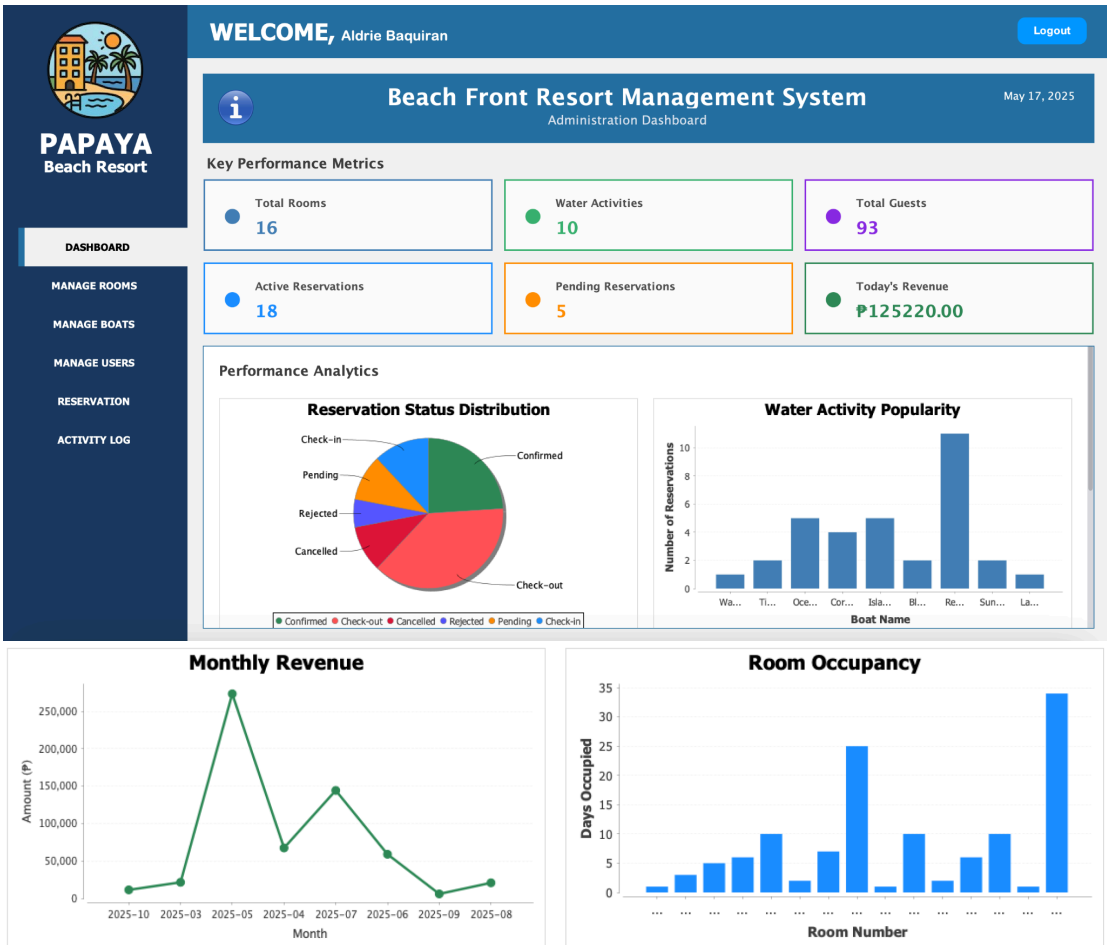
Date Created	Reservation ID	Guest Name	Check-In	Check-Out	Total
2025-04-27 00:31:22	RES-20250426-001	Aldrie Baquiran	2025-04-25	2025-04-26	P8640.00
2025-04-27 00:48:35	RES-20250426-002	test test	2025-04-25	2025-04-26	P5020.00
2025-04-27 00:52:56	RES-20250426-003		2025-04-25	2025-04-26	P7820.00
2025-04-27 07:17:30	RES-20250426-005	v v	2025-04-25	2025-04-26	P5320.00
2025-04-27 07:18:00	RES-20250426-006	c v	2025-04-28	2025-04-29	P8520.00
2025-04-28 02:32:49	RES-20250427-005	b b	2025-05-01	2025-05-03	P5120.00
2025-05-03 06:54:19	RES-20250502-002	Aldrich Baquiran	2025-05-15	2025-05-17	P9240.00
2025-05-10 18:10:57	RES-20250510-001	c c	2025-05-09	2025-05-16	P35120.00
2025-05-10 23:39:08	RES-20250510-002	Ryan Palo	2025-05-09	2025-05-12	P20840.00
2025-05-11 03:07:08	RES-20250510-003	Ivan Cabatian	2025-05-10	2025-05-16	P42120.00
2025-05-11 03:07:59	RES-20250510-004	Ivan Cabatian	2025-05-29	2025-05-30	P5120.00
2025-05-11 03:12:54	RES-20250510-006	Boiser Guevarra	2025-05-28	2025-05-30	P14140.00
2025-05-11 04:14:28	RES-20250510-007	Paulo Boiser	2025-06-10	2025-06-13	P11640.00
2025-05-11 08:16:10	RES-20250510-008	Eloisa Tercias	2025-05-10	2025-05-11	P12040.00
2025-05-12 12:27:56	RES-20250511-004	Emily Tercias	2025-09-19	2025-09-20	P5940.00
2025-05-13 03:54:06	RES-20250512-003	Claire Cabral	2025-05-20	2025-05-21	P7720.00
2025-05-14 01:36:57	RES-20250513-003	Jasmin Baquiran	2025-05-14	2025-05-15	P3620.00

A record of all reservations that have been successfully completed. Useful for report generation and auditing.



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Admin Interface



This admin panel for PAPAYA Beach Resort offers a quick overview of key metrics like total rooms, guests, revenue, and reservation statuses, alongside charts visualizing reservation distribution, activity popularity, monthly revenue trends, and room occupancy.

List of Boats					Add New Boat	Update Details	Delete Boat	View Reservation
All	Search here...							
Boat Number	Name	Details	Capacity	Price/Tour				
BN-001	Island Hopper	Perfect for small island tours.	6	1500.0				
BN-002	Sunset Cruiser	Ideal for sunset cruises and private trips.	8	2000.0				
BN-003	Wave Rider	Fast boat for adventure seekers.	4	1800.0				
BN-004	Coral Explorer	Great for snorkeling and coral viewing.	10	2500.0				
BN-005	Sea Breeze	Comfortable boat for families.	7	1700.0				

Admins can add new boats, update pricing, and view their booking schedules.



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List of Rooms						Add New Room	Update Details	Delete Room	View Reservation
All	Search here...								
Room Number	Room Image	Type	Description	Max Occupancy	Price/Night				
R001		Deluxe Room	A spacious room with a queen-size bed, ...	2	3500.0				
R002		Family Suite	Large suite with two double beds, sofa, ...	6	5000.0				
R003		Beachfront Villa	Direct access to the beach, king-size bed...	2	7000.0				

Admin can add, update, or delete room records. The panel also shows each room’s reservation schedule to help avoid conflicts.

List of Users					Add New Users	Update Details	Delete Users
All	Search here...						
#	Role	Full Name	Phone	Email			
1	Guest	Claire Cabral	09123456780	claire@gmail.com			
2	Admin	Aldrie Baquiran	09000000002	admin@gmail.com			
3	Staff	Clarisse Cabral	09000000003	clarisse@gmail.com			
4	Guest	Emily Cruz	09123456780	emily@gmail.com			

This panel allows admins to add or manage system users. They can register new guests, staff, or even other admins.

Reservation History							
All	Search here...						
All							
Pending							
Rejected							
Confirmed							
Check-in							
Check-out							
2025-05-17 08:55:12	RES-20250516-002	Aldrie Baquiran	2025-06-17	2025-06-18	P8860.00	Rejected	
2025-05-17 09:02:03	RES-20250516-003	Aldrie Baquiran	2025-05-22	2025-05-23	P5240.00	Confirmed	

A complete log of all reservations ever made. Admins can filter records by date, guest, or status for reporting and review.



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Activity Log

All

Search here...

Log ID	Full Name	Role	Action Type	Description	Timestamp
253	Clarisse Cabral	Staff	UPDATE_RESERVATION_STATUS	View	2025-05-17 16:38:19.0
252	Clarisse Cabral	Staff	UPDATE_RESERVATION_STATUS	View	2025-05-17 16:38:12.0
251	Clarisse Cabral	Staff	UPDATE_RESERVATION_STATUS	View	2025-05-17 16:38:06.0
250	Clarisse Cabral	Staff	UPDATE_RESERVATION_STATUS	View	2025-05-17 16:37:59.0
249	Clarisse Cabral	Staff	RESERVATION	View	2025-05-17 16:37:45.0
248	Clarisse Cabral	Staff	RESERVATION	View	2025-05-17 16:37:39.0
247	Clarisse Cabral	Staff	RESERVATION	View	2025-05-17 16:37:27.0
246	Clarisse Cabral	Staff	RESERVATION	View	2025-05-17 16:37:24.0
245	Clarisse Cabral	Staff	CONFIRM_RESERVATION	View	2025-05-17 16:37:21.0
244	Clarisse Cabral	Staff	CONFIRM_RESERVATION	View	2025-05-17 16:37:19.0
243	Clarisse Cabral	Staff	CONFIRM_RESERVATION	View	2025-05-17 16:37:14.0
242	Clarisse Cabral	Staff	CONFIRM_RESERVATION	View	2025-05-17 16:37:11.0
241	Clarisse Cabral	Staff	LOGIN	View	2025-05-17 16:37:01.0
240	Chame Tapang	Guest	LOGOUT	View	2025-05-17 16:36:45.0
239	Chame Tapang	Guest	RESERVATION	View	2025-05-17 16:35:53.0
238	Chame Tapang	Guest	LOGIN	View	2025-05-17 16:34:49.0
237	Chame Tapang	Guest	Create Account	View	2025-05-17 16:34:38.0

The "Activity Log" tracks user actions with details like user, role, action type, description, and timestamp, allowing administrators to monitor system activity. Selecting an entry may show further details.

VII. Insights

A database-driven reservation system empowers Papaya Beach Resort with real-time visibility of their operations. It reduces overbooking, improves payment tracking, and enhances guest satisfaction through a smoother booking experience. Staff and admin benefit from faster data access and better coordination.

VIII. Challenges and Learnings

One of the key challenges faced during the development of the system was syncing the activity time slots with the guest's available stay dates, ensuring that guests could only book boat tours within their check-in and check-out period. Another difficulty involved accurately computing the 30% downpayment from the total cost and validating payment entries, especially across multiple payment methods. Additionally, implementing secure login functionality across



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different user roles—Guest, Staff, and Admin—required careful handling of authentication and role-based access control.

Throughout the project, I learned that using database triggers was crucial in maintaining data integrity, especially for automating status updates and ensuring consistent reservation records. Writing and optimizing structured SQL queries greatly improved the system’s performance and responsiveness. Lastly, conducting real-world testing scenarios was essential to identify usability issues early and ensure that the system operates reliably under actual usage conditions.

IX. SQL Query Statement used

- **SQL Tables**

User Management Tables

```
CREATE TABLE user_details (  
    user_id INT AUTO_INCREMENT PRIMARY KEY,  
    role ENUM('Guest', 'Staff', 'Admin'),  
    full_name VARCHAR(255),  
    phone VARCHAR(15),  
    email VARCHAR(255) UNIQUE,  
    password VARCHAR(255)  
);  
  
CREATE TABLE guest (  
    guest_id INT AUTO_INCREMENT PRIMARY KEY,  
    user_id INT,  
    guest_name VARCHAR(255),  
    email VARCHAR(255),  
    contact VARCHAR(15),  
    address VARCHAR(255),  
    FOREIGN KEY (user_id) REFERENCES user_details(user_id)  
    -- Stores detailed profile information of the guest  
);
```

Room Management Tables

```
CREATE TABLE room (  
    room_id INT AUTO_INCREMENT PRIMARY KEY,  
    room_number VARCHAR(10),  
    room_image LONGBLOB,  
    room_type VARCHAR(50),  
    room_price DECIMAL(10,2),  
    description TEXT,  
    max_occupancy INT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
    -- Contains details about each resort room  
);
```

Reservation Management Tables



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```
CREATE TABLE room_reservation (  
    room_reservation_id INT AUTO_INCREMENT PRIMARY KEY,  
    reservation_number VARCHAR(255) UNIQUE,  
    user_id INT,  
    guest_id INT,  
    room_number VARCHAR(255),  
    adult INT,  
    child INT,  
    total_guests INT,  
    check_in_date DATE,  
    check_out_date DATE,  
    total_room_price DECIMAL(10,2),  
    total_entrance_fee DECIMAL(10,2),  
    total_ecological_fee DECIMAL(10,2),  
    boat_tour_status ENUM('Availed', 'Not Availed'),  
    status ENUM('Reserved', 'Completed'),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (user_id) REFERENCES user_details(user_id),  
    FOREIGN KEY (guest_id) REFERENCES guest(guest_id),  
    FOREIGN KEY (room_number) REFERENCES room(room_number)  
    -- Tracks specific room bookings and associated guest details  
);  
  
CREATE TABLE reservation (  
    reservation_id INT AUTO_INCREMENT PRIMARY KEY,  
    reservation_number VARCHAR(255) UNIQUE,  
    user_id INT,  
    guest_id INT,  
    room_reservation_id INT,  
    check_in_date DATE,  
    check_out_date DATE,  
    total_price DECIMAL(10,2),  
    status ENUM('Pending', 'Confirmed', 'Check-in', 'Check-out', 'Cancelled')  
    DEFAULT 'Pending',  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (guest_id) REFERENCES guest(guest_id),  
    FOREIGN KEY (room_reservation_id) REFERENCES  
    room_reservation(room_reservation_id)  
    -- General reservation record combining room and guest data  
);
```

Payment Management Tables

```
CREATE TABLE payment (  
    payment_id INT AUTO_INCREMENT PRIMARY KEY,  
    reservation_id INT,  
    payment_type ENUM('Downpayment', 'Balance'),  
    payment_method ENUM('GCash', 'Maya', 'PayPal', 'On Site'),  
    amount DECIMAL(10,2),  
    payment_date DATETIME DEFAULT CURRENT_TIMESTAMP,  
    reference_number VARCHAR(100),  
    status ENUM('Pending', 'Paid', 'Failed') DEFAULT 'Pending',
```



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```
FOREIGN KEY (reservation_id) REFERENCES reservation(reservation_id)
-- Stores payment transactions and status for each reservation
```

```
);
```

Boat Tour Management Tables

```
CREATE TABLE boat (
    boat_id INT AUTO_INCREMENT PRIMARY KEY,
    boat_number VARCHAR(20),
    boat_name VARCHAR(255),
    description TEXT,
    capacity INT,
    registration_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    tour_price DECIMAL(10,2)
    -- Stores details of boats available for reservation
);
```

```
CREATE TABLE boat_reservation (
    boat_reservation_id INT AUTO_INCREMENT PRIMARY KEY,
    room_reservation_id INT,
    guest_id INT,
    boat_id INT,
    boat_tour_date DATE,
    boat_tour_start_time TIME,
    boat_tour_end_time TIME,
    tour_price DECIMAL(10,2) DEFAULT 0.00,
    status ENUM('Reserved', 'Completed'),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (room_reservation_id) REFERENCES
room_reservation(room_reservation_id),
    FOREIGN KEY (guest_id) REFERENCES guest(guest_id),
    FOREIGN KEY (boat_id) REFERENCES boat(boat_id)
    -- Tracks scheduled boat tours linked to room reservations
);
```

Activity Log Table

```
CREATE TABLE activity_log (
    log_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    action_type VARCHAR(50),
    action_description TEXT,
    action_timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES user_details(user_id)
);
```

Reservation Rating Tables

```
CREATE TABLE reservation_ratings (
    rating_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    room_reservation_id INT,
    rating_value INT CHECK (rating_value BETWEEN 1 AND 5),
```



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```
comments TEXT,  
rating_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (user_id) REFERENCES user_details(user_id),  
FOREIGN KEY (room_reservation_id) REFERENCES  
room_reservation(room_reservation_id)  
);
```

• SQL Queries

Authentication Queries

```
-- Login Query  
SELECT full_name, role, user_id  
FROM user_details  
WHERE email = ? AND password = ?  
LIMIT 1;
```

```
-- User Registration  
INSERT INTO user_details (full_name, phone, email, password, role)  
VALUES (?, ?, ?, ?, 'Guest');
```

```
-- Select all users with full details  
SELECT user_id, role, full_name, phone, email, password  
FROM user_details;
```

Room Management Queries

```
-- Select all rooms  
SELECT * FROM room;
```

```
-- Available Rooms by Guest Count and Date Range  
SELECT r.room_number, r.room_type, r.max_occupancy, r.room_price  
FROM room r  
WHERE r.max_occupancy >= ?  
AND r.room_number NOT IN (  
    SELECT room_number  
    FROM room_reservation  
    WHERE status = 'Reserved'  
    AND (? < check_out_date AND ? > check_in_date)  
)  
ORDER BY r.room_price ASC;
```

```
-- Available Rooms (Detailed)  
SELECT r.room_number, r.room_type, r.description, r.room_price, r.room_image,  
r.max_occupancy  
FROM room r  
WHERE r.max_occupancy >= ?  
AND r.room_number NOT IN (  
    SELECT room_number  
    FROM room_reservation  
    WHERE status = 'Reserved'  
    AND (? <= check_out_date AND ? >= check_in_date)  
)  
ORDER BY r.room_price ASC;
```



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Boat Management Queries

```
-- Select all boats
SELECT * FROM boat;

-- Available Boats by Capacity and Time Overlap
SELECT b.boat_name, b.description, b.tour_price
FROM boat b
WHERE b.capacity >= ?
AND b.boat_id NOT IN (
    SELECT br.boat_id
    FROM boat_reservation br
    WHERE br.status = 'Reserved'
    AND br.boat_tour_date = ?
    AND (? < br.boat_tour_end_time AND ? > br.boat_tour_start_time)
)
ORDER BY b.tour_price ASC;

-- Select reserved boat reservations with room reservation info
SELECT br.boat_reservation_id, rr.reservation_number, br.boat_tour_date,
       br.boat_tour_start_time, br.boat_tour_end_time
FROM boat_reservation br
JOIN room_reservation rr ON br.room_reservation_id = rr.room_reservation_id
WHERE br.status = 'Reserved';
```

Reservation Management Queries

```
-- Insert Guest Info
INSERT INTO guest (user_id, guest_name, email, contact, address)
VALUES (?, ?, ?, ?, ?);

-- Insert Room Reservation
INSERT INTO room_reservation (
    reservation_number, user_id, guest_id, room_number, adult, child,
    total_guests, check_in_date, check_out_date, total_room_price,
    total_entrance_fee, total_ecological_fee, boat_tour_status, status,
    created_at
)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, NOW());

-- Insert Boat Reservation
INSERT INTO boat_reservation (
    room_reservation_id, guest_id, boat_id, boat_tour_date,
    boat_tour_start_time, boat_tour_end_time, tour_price, status, created_at
)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, NOW());

-- Insert General Reservation Record
INSERT INTO reservation (
    reservation_number, guest_id, user_id, room_reservation_id,
    check_in_date, check_out_date, total_price, status, created_at
)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, NOW());
```



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```
-- Select reserved room reservations
SELECT reservation_number, check_in_date, check_out_date
FROM room_reservation
WHERE status = 'Reserved';

-- Get Pending Reservations for a User
SELECT r.reservation_number, r.check_in_date, rr.room_reservation_id,
       r.check_out_date, rr.room_number, rm.room_image
FROM reservation r
JOIN room_reservation rr ON r.reservation_number = rr.reservation_number
JOIN room rm ON rr.room_number = rm.room_number
WHERE r.user_id = ? AND r.status = 'Pending';

-- Get Non-Pending Reservations for a User
SELECT r.reservation_number, r.check_in_date, rr.room_reservation_id,
       r.check_out_date, rr.room_number, rm.room_image, rm.room_type
FROM reservation r
JOIN room_reservation rr ON r.reservation_number = rr.reservation_number
JOIN room rm ON rr.room_number = rm.room_number
WHERE r.user_id = ? AND r.status != 'Pending';

-- Select reservation details with guest names
SELECT reservation_number, guest.guest_name, r.check_in_date,
       r.check_out_date, r.total_price, r.status, r.created_at
FROM reservation r
JOIN guest ON r.guest_id = guest.guest_id;

-- Select pending reservations with guest info
SELECT reservation_number, guest.guest_name, r.check_in_date, r.check_out_date,
       r.total_price, r.created_at, r.reservation_id
FROM reservation r
JOIN guest ON r.guest_id = guest.guest_id
WHERE r.status = 'Pending';

-- Parameterized version of the above (for PreparedStatement)
SELECT reservation_number, guest.guest_name, r.check_in_date, r.check_out_date,
       r.total_price, r.created_at, r.reservation_id
FROM reservation r
JOIN guest ON r.guest_id = guest.guest_id
WHERE r.status = ?;

-- Get reservation info by reservation number
SELECT r.*, g.guest_name
FROM reservation r
JOIN guest g ON r.guest_id = g.guest_id
WHERE r.reservation_number = ?;

-- Join room and boat reservations by room_reservation_id
SELECT rr.*, br.boat_id, br.boat_tour_date, br.boat_tour_start_time,
```



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```
br.boat_tour_end_time, br.tour_price
FROM room_reservation rr
LEFT JOIN boat_reservation br ON rr.room_reservation_id =
br.room_reservation_id
WHERE rr.room_reservation_id = ?;
```

```
-- Update reservation status
UPDATE reservation SET status = ?
WHERE reservation_number = ?;
```

Payment Processing Queries

```
-- Insert Payment Info
INSERT INTO payment (
    reservation_id, payment_type, payment_method, amount, reference_number,
    status
)
VALUES (?, ?, ?, ?, ?, ?);
```

```
-- Query to get balance and amount paid
SELECT r.total_price, COALESCE(SUM(p.amount), 0) AS paid
FROM reservation r
LEFT JOIN payment p ON r.reservation_id = p.reservation_id AND p.status =
'Paid'
WHERE r.reservation_id = ?;
```

```
-- Update payment status
UPDATE payment SET amount = ?, status = 'Paid'
WHERE payment_id = ?;
```

```
-- Insert a new balance payment
INSERT INTO payment (reservation_id, payment_type, payment_method, amount,
status)
VALUES (?, 'Balance', 'On Site', ?, 'Paid');
```

• Database Triggers

```
-- Trigger to prevent overlapping boat reservations
BEGIN
    IF EXISTS (
        SELECT 1
        FROM boat_reservation
        WHERE boat_id = NEW.boat_id
        AND boat_tour_date = NEW.boat_tour_date
        AND (
            (NEW.boat_tour_start_time BETWEEN boat_tour_start_time AND
boat_tour_end_time)
            OR
            (NEW.boat_tour_end_time BETWEEN boat_tour_start_time AND
boat_tour_end_time)
            OR
            (boat_tour_start_time BETWEEN NEW.boat_tour_start_time AND
NEW.boat_tour_end_time)
```




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```
)  
) THEN  
    SIGNAL SQLSTATE '45000'  
    SET MESSAGE_TEXT = 'Overlapping reservation for this boat already exists  
on the selected time.';  
END IF;  
END;  
  
-- Trigger to prevent overlapping room reservations  
BEGIN  
    IF EXISTS (  
        SELECT 1 FROM room_reservation  
        WHERE room_number = NEW.room_number  
        AND (  
            (NEW.check_in_date BETWEEN check_in_date AND DATE_SUB(check_out_date,  
INTERVAL 1 DAY)) OR  
            (NEW.check_out_date BETWEEN DATE_ADD(check_in_date, INTERVAL 1 DAY) AND  
check_out_date) OR  
            (NEW.check_in_date <= check_in_date AND NEW.check_out_date >=  
check_out_date)  
        )  
        AND status = 'Reserved'  
    ) THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT = 'This room is already reserved during the selected  
date range.';  
    END IF;  
END;
```

● Database Views

```
-- View: Upcoming confirmed reservations  
CREATE VIEW upcoming_confirmed_reservations AS  
SELECT * FROM reservation  
WHERE status = 'confirmed' AND reservation_date >= CURDATE();  
  
-- View: Count of cancelled reservations  
CREATE VIEW view_count_cancelled_reservations AS  
SELECT COUNT(*) AS cancelled_count FROM reservation  
WHERE status = 'cancelled';  
  
-- View: Count of confirmed reservations  
CREATE VIEW view_count_confirmed_reservations AS  
SELECT COUNT(*) AS confirmed_count FROM reservation  
WHERE status = 'confirmed';  
  
-- View: Count of pending reservations  
CREATE VIEW view_count_pending_reservations AS  
SELECT COUNT(*) AS pending_count FROM reservation  
WHERE status = 'pending';  
  
-- View: Guest users only
```



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College of Informatics and Computing Sciences

```
CREATE VIEW view_guest_users AS
SELECT * FROM user_details
WHERE role = 'guest';

-- View: All pending reservations
CREATE VIEW view_pending_reservations AS
SELECT * FROM reservation
WHERE status = 'pending';

-- View: Total number of boats
CREATE VIEW view_total_boats AS
SELECT COUNT(*) AS total_boats FROM boat;

-- View: Total number of boat reservations
CREATE VIEW view_total_boat_reservations AS
SELECT COUNT(*) AS total_boat_reservations FROM boat_reservation;

-- View: Total number of rooms
CREATE VIEW view_total_rooms AS
SELECT COUNT(*) AS total_rooms FROM room;

-- View: Total number of room reservations
CREATE VIEW view_total_room_reservations AS
SELECT COUNT(*) AS total_room_reservations FROM room_reservation;
```